

No largest operator ideal with Hilbert identity spaces

A candidate full negative answer to Question 6.2 of arXiv:2005.07672

Automated mathematics run `fa_banach_001`, lane 5

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Status. Candidate full solution, likely valid, requiring expert human review. The argument answers the question negatively over both real and complex scalars.

1 Source question

Ferenczi's paper [1] asks on page 22, Question 6.2, whether there is a largest operator ideal whose identity spaces are exactly the separable Hilbert spaces. Here is the source passage.

6. Comments and problems

Our results leave open the following new version of [27] Problem 2.2.8.

QUESTION 6.1. *For which space ideals A does there exist a largest operator ideal U with $A = \text{Space}(U)$?*

Recall that a space ideal is a class of spaces containing the finite dimensional ones and stable under taking direct sums and complemented subspaces. For any ideal U , $\text{Space}(U)$ is a space ideal, [27] Theorem 2.1.3, and conversely a space ideal A always coincides with $\text{Space}(\text{Op}(A))$, [27] Theorem 2.2.5. And our main result is that the answer to Question [6.1] is negative for the space ideal F of finite dimensional spaces.

The "next" natural example seems to be the space ideal H of separable spaces isomorphic to a Hilbert space (finite or infinite dimensional). So it seems natural to ask:

QUESTION 6.2. *Does there exist a largest operator ideal U with $H = \text{Space}(U)$?*

Pietsch ([27] 4.3.7) considered a natural candidate for the largest operator ideal V such that $\text{Space}(V) = \text{Space}(U)$; namely U^{rad} , defined by

$$S \in U^{\text{rad}}(X, Y) \Leftrightarrow \forall L \in L(Y, X), W(I_X - LS) = I_X - T$$

Figure 1: Question 6.2 and its context in [1, p. 22].

Writing $\mathcal{H}$ for the space ideal of separable Banach spaces isomorphic to a Hilbert space, the question is:

Does there exist a largest operator ideal $\mathcal{U}$ with $\mathcal{H} = \text{Space}(\mathcal{U})$?

2 Definitions and imported facts

Fix one scalar field $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and put $H = \ell_2(\mathbb{K})$. For a Banach space E , let $\text{Op}^{<\omega}(E)$ be the operator ideal consisting of all operators which factor through E^n for some $n \geq 1$. If $\mathcal{U}$ is an

operator ideal, then

$$\text{Space}(\mathcal{U}) = \{Z : \text{Id}_Z \in \mathcal{U}\}.$$

We use four established inputs.

- (i) Gowers and Maurey's shift space $X = X_S(\mathbb{K})$ contains no unconditional basic sequence [2, Section 5.2]. In particular, it contains no infinite-dimensional Hilbert space.
- (ii) Ferenczi constructs, for $t \in \{1, -1\}$, a compact operator K_t such that

$$T_t = \text{Id}_X - tS + K_t, \quad Y_t = \text{ran } T_t,$$

where Y_t has infinite codimension in X [1, Section 4]. Moreover, if $m, n \geq 1$, then no infinite-dimensional complemented subspace of X^m is isomorphic to a subspace of Y_t^n [1, Proposition 4.4].

- (iii) If E and F are essentially incomparable, then every complemented subspace of $E \oplus F$ is isomorphic to $E_0 \oplus F_0$ for complemented subspaces $E_0 \subseteq E$ and $F_0 \subseteq F$ [3].
- (iv) Strictly singular perturbations preserve the upper semi-Fredholm class. Equivalently, if A is upper semi-Fredholm and R is strictly singular, then $A + R$ is upper semi-Fredholm [4, Chapter 2].

Recall also that strictly singular operators are inessential. Thus, if every operator in both directions between two spaces is strictly singular, those spaces are essentially incomparable.

3 Proof intuition

Ferenczi destroys a hypothetical largest *proper* ideal using two factorization ideals associated with X and with Y_t . To preserve exactly the Hilbert identity spaces, adjoin a common Hilbert summand before repeating the construction:

$$\mathcal{J}_t = \text{Op}^{<\omega}(H \oplus X) \cap \text{Op}^{<\omega}(H \oplus Y_t).$$

The common H makes every Hilbert identity belong to $\mathcal{J}_t$. The point is that it adds no other identity. Indeed, a space whose identity belongs to $\mathcal{J}_t$ is complemented in both $H \oplus X^m$ and $H \oplus Y_t^n$. Essential incomparability separates each complemented copy into a Hilbert piece and an exotic piece. Comparing the two decompositions forces the exotic piece, modulo a finite-dimensional kernel, to embed from X^m into Y_t^n . Ferenczi's rigidity theorem rules this out. Hence the common identity spaces are exactly Hilbertian, while the two operators T_1, T_{-1} still belong to the corresponding ideals and still have a Fredholm sum.

4 The common-complemented-subspace argument

Lemma 1. *For every $m, n \geq 1$, neither X^m nor Y_t^n contains a subspace isomorphic to H .*

Proof. The space X contains no unconditional basic sequence, hence no copy of H . Suppose H embedded into X^m . Each coordinate of the embedding is strictly singular: otherwise it would be bounded below on an infinite-dimensional subspace of H , and every such subspace is isomorphic to H , producing a copy of H in X . The column embedding, being a finite sum of strictly singular operators after the coordinate inclusions, would then be strictly singular, contradicting that it is bounded below. Thus X^m contains no H . Since Y_t^n is a subspace of X^n , the second assertion follows. $\square$

Lemma 2. *For each $m, n \geq 1$, the pairs (H, X^m) and (H, Y_t^n) are essentially incomparable. Consequently every complemented subspace of $H \oplus X^m$ is isomorphic to $H_0 \oplus E_0$, where H_0 is Hilbertian and E_0 is complemented in X^m ; the analogous assertion holds for $H \oplus Y_t^n$.*

Proof. If an operator $A : H \rightarrow X^m$ were not strictly singular, it would be bounded below on an infinite-dimensional subspace of H , hence would put a copy of H in X^m , contrary to Lemma 1. If $B : X^m \rightarrow H$ were not strictly singular, the subspace on which it is bounded below would itself be Hilbertian, giving the same contradiction. Thus all cross operators are strictly singular and therefore inessential. The same argument applies to Y_t^n . The stated decompositions now follow from Gonzalez's theorem [3]. $\square$

Lemma 3. *If a Banach space Z is isomorphic to a complemented subspace of both $H \oplus X^m$ and $H \oplus Y_t^n$, then Z is Hilbertian.*

Proof. By Lemma 2, there are decompositions

$$Z \simeq H_0 \oplus E_0 \simeq H_1 \oplus F_0,$$

where H_0, H_1 are Hilbertian, E_0 is complemented in X^m , and F_0 is complemented in Y_t^n . Let $R : H_0 \oplus E_0 \rightarrow H_1 \oplus F_0$ be an isomorphism. On E_0 , write

$$R|_{E_0} = (A, B), \quad A : E_0 \rightarrow H_1, \quad B : E_0 \rightarrow F_0.$$

The operator A is strictly singular. Indeed, otherwise A would be bounded below on an infinite-dimensional subspace of E_0 , making that subspace Hilbertian and contradicting Lemma 1.

The column $(A, B) : E_0 \rightarrow H_1 \oplus F_0$ is bounded below, hence upper semi-Fredholm. It is a strictly singular perturbation of $(0, B)$, so $(0, B)$, and equivalently B , is upper semi-Fredholm. Therefore B has finite dimensional kernel and closed range. Choose a closed complement E_1 of $\ker B$ in E_0 . Then E_1 is finite codimensional in E_0 , is complemented in X^m , and $B|_{E_1}$ is an isomorphic embedding into $F_0 \subseteq Y_t^n$. Ferenczi's Proposition 4.4 forces E_1 to be finite dimensional. Hence E_0 is finite dimensional, and $Z \simeq H_0 \oplus E_0$ is Hilbertian. $\square$

5 Full negative answer

Theorem 4. *Over either the real or complex scalars, there is no largest operator ideal $\mathcal{U}$ satisfying $\text{Space}(\mathcal{U}) = \mathcal{H}$.*

Proof. For $t \in \{1, -1\}$ define

$$\mathcal{J}_t := \text{Op}^{<\omega}(H \oplus X) \cap \text{Op}^{<\omega}(H \oplus Y_t).$$

We first show

$$\text{Space}(\mathcal{J}_t) = \mathcal{H}. \tag{1}$$

Every separable Hilbert space is finite dimensional or isomorphic to H , so its identity factors through the common H summand and belongs to $\mathcal{J}_t$. Conversely, suppose $\text{Id}_Z \in \mathcal{J}_t$. A factorization of an identity through a space realizes its domain as a complemented subspace of that space. Hence, for some $m, n \geq 1$, the space Z is isomorphic to a complemented subspace of both

$$(H \oplus X)^m \simeq H \oplus X^m \quad \text{and} \quad (H \oplus Y_t)^n \simeq H \oplus Y_t^n.$$

Lemma 3 says that Z is Hilbertian. This proves (1).

Assume now that a largest ideal $\mathcal{U}$ with $\text{Space}(\mathcal{U}) = \mathcal{H}$ exists. By (1), it contains both $\mathcal{J}_1$ and $\mathcal{J}_{-1}$. The operator $T_t : X \rightarrow Y_t$, also viewed as an operator $X \rightarrow X$, factors through X and

through Y_t . It therefore factors through both $H \oplus X$ and $H \oplus Y_t$, so $T_t \in \mathcal{J}_t$. Consequently $T_1, T_{-1} \in \mathcal{U}$ and

$$F := T_1 + T_{-1} = 2\text{Id}_X + K_1 + K_{-1} \in \mathcal{U}(X, X).$$

The operator F is Fredholm. Choose a parametrix R with $RF = \text{Id}_X - Q$ for a finite-rank operator Q . Ideality gives $RF \in \mathcal{U}$, and every operator ideal contains Q ; hence $\text{Id}_X = RF + Q \in \mathcal{U}$. Thus $X \in \text{Space}(\mathcal{U}) = \mathcal{H}$, impossible because X contains no unconditional basic sequence and so is not Hilbertian. This contradiction proves the theorem. $\square$

6 Verification and limitations

The proof is non-computational. The exact source question, the construction of Y_t , Proposition 4.4, and the Fredholm identity for $T_1 + T_{-1}$ were checked against the source TeX and PDF. The no-unconditional-basic-sequence statement was checked in the primary Gowers–Maurey source [2, Section 5.2], and the direct-sum theorem against Gonzalez’s article record and abstract [3].

A bounded novelty search on 9 August 2026 covered the run registry, cheap arXiv source indexes, and indexed web/arXiv results for the exact question and the phrases “largest operator ideal”, “Hilbert space ideal”, and “ $\text{Space}(\mathcal{U}) = \mathcal{H}$ ”. It found the source paper and later citations of its proper-ideal theorem, but no claimed resolution of Question 6.2. This is not exhaustive, so novelty confidence is moderate. The result should be cited only as a candidate full solution until an expert independently checks the three imported structural facts, especially the semi-Fredholm perturbation step in Lemma 3.

References

- [1] V. Ferenczi, *There is no largest proper operator ideal*, arXiv:2005.07672v2; Math. Ann. 387 (2023), 1043–1072.
- [2] W. T. Gowers and B. Maurey, *Banach spaces with small spaces of operators*, Math. Ann. 307 (1997), 543–568; arXiv:math/9407209.
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